

TrES-5b

R-1253

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_211014 · created 2026-07-31T12:00:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459501.6388 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.109 +/- 0.044
Transit depth (Rp/Rs)^2	1.18 +/- 0.96 [%]
Orbital Inclination (inc)	82.23 +/- 2.64
Airmass coefficient 1 (a1)	1.367 +/- 0.03
Airmass coefficient 2 (a2)	-0.255 +/- 0.049
Scatter in the residuals of the lightcurve fit is	2.2 %
Best Comparison Star	#2 - [163, 115]
Optimal Method	PSF photometry
Transit Duration (day)	0.049 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.109 ± 0.044 vs 0.142 (deviation 0.75σ)
Duration fit vs published	0.049 d vs 0.0758 d (deviation 0.99σ)
Mid-time O-C	-8.1 min
Depth SNR	2.5
Residual scatter	2.2 %

Light curve

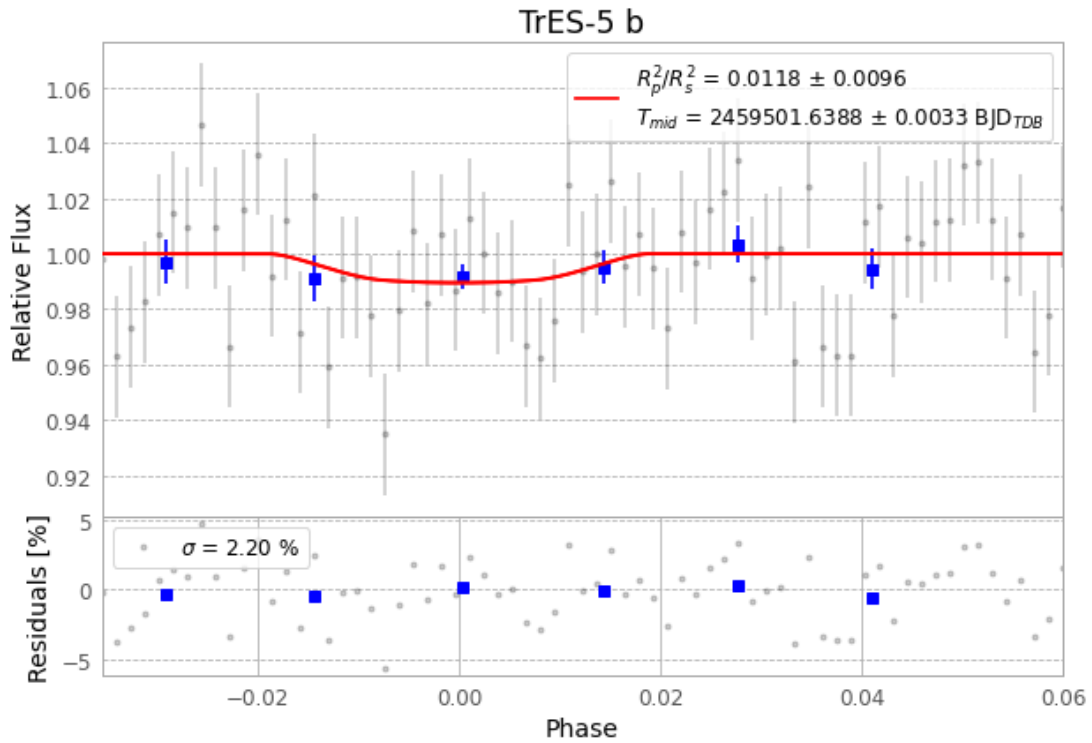


Figure 1 — FinalLightCurve_TrES-5 b_2021-10-13.png (SHA-256 2cdf0b8489875468...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [456, 266], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2fcd4d72d5cfb9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), TRES-5211014020012.FITS → TRES-5211014052412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-211014.log (99c06019a8d8...), FinalLightCurve_TrES-5 b_2021-10-13.png (2cdf0b848987...), FinalLightCurve_TrES-5 b_2021-10-13.csv (5b7e6b7f65ce...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:00Z.